6 Nested Conditionals: Decision Trees

AGES 8-12

MASCOT GOES HERE

KID INSTRUCTIONS

Last activity, you used one Yes/No question to make a decision. But real life is messier — most decisions have two or three questions stacked together. That's called a **decision tree**!

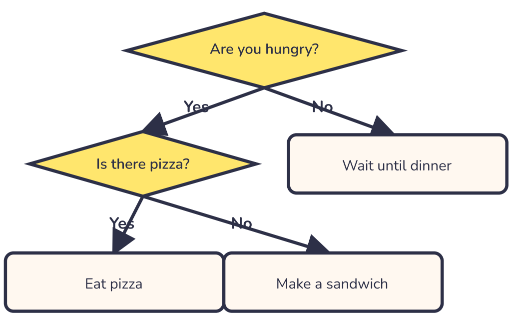
Nested

Nested — When something is INSIDE another thing of the same type. Like a smaller box inside a bigger box, or a question inside another question.
Example: Just like nested loops (Activity 4!), conditionals can live inside other conditionals. A question inside a question = a nested conditional.

Basic Version

Follow the Decision Tree — Each tree has TWO questions. Follow the path from top to bottom, then answer the questions below!

Tree 1 — Lunch Decision



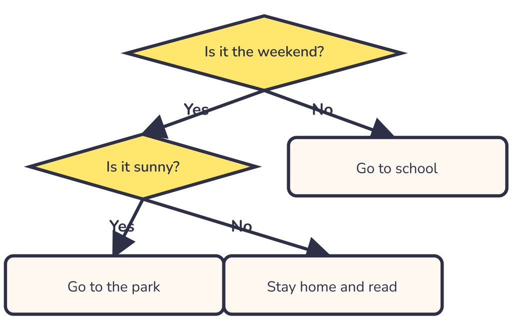
Question 1: It's noon, you're starving, and there's leftover pizza in the fridge. What do you do?

→ ______

Question 2: You're not hungry yet. What happens?

→ ______

Tree 2 — Weekend Plan



Question 1: It's a sunny Saturday. What do you do?

→ ______

Question 2: It's Tuesday. What do you do?

→ ______

Challenge Version

PART A — Write the Nested Conditional in Shorthand

```
IF [outer condition] THEN
   IF [inner condition] THEN [action] OTHERWISE [other
action]
OTHERWISE [other action]
```

Example: A movie ticket: kids under 12 get $5 tickets, kids 12+ pay $10. But if it's Tuesday, everyone pays $5.

```
IF it's Tuesday THEN charge $5
OTHERWISE
   IF the customer is under 12 THEN charge $5
   OTHERWISE charge $10
```

Scenario A1

A school dismissal — If it's raining, students wait in the cafeteria. Otherwise, if it's a school day, they go outside to be picked up. If it's not a school day, they don't come to school in the first place.

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______

Scenario A2

A library book return — If a book is overdue, the kid pays a fine. If it's not overdue and the book is damaged, the kid pays a damage fee. If it's not overdue and not damaged, no charge.

______

______

______

______

PART B — Spot the Bug

A programmer wrote this nested conditional for walking a pet dog. There's a bug in it!

```
IF the weather is good THEN walk the dog outside
OTHERWISE
   IF it's raining THEN walk the dog inside
   OTHERWISE walk the dog outside      ← something's off here
```

What's the bug? How would you fix it?

______

______

______

Self-Check

- [ ] I followed both decision trees and answered all 4 questions
- [ ] I wrote 2 nested conditionals in IF-THEN-OTHERWISE shorthand
- [ ] I found and fixed the bug in the dog-walking conditional
- [ ] I can explain in my own words what "nested" means

Reflection Box

Think of a decision you made today that had MORE than one Yes/No question. Write it down and label which question came first.

Bonus: Design Your Own Decision Tree

Pick ONE everyday decision you make often. Then build a decision tree with TWO Yes/No questions to handle it.

Pick a starter (or invent your own):

- What do I wear today?
- What do I eat for breakfast?
- Should I play outside?

Draw your decision tree here. Use diamonds for questions, arrows for Yes/No, and boxes for the actions.

Try it! Describe a situation that ends at one of your action boxes. Which box did you land on?

TEACHER / PARENT NOTE

This activity introduces nested conditionals — also called decision trees or "if-elif-else" in real code. Kids learn that real decisions aren't usually one Yes/No question but a chain of them, which mirrors how real software (game logic, app permissions, sorting algorithms) works. Discussion prompt: Ask your child to spot a real nested conditional in your family's daily routine (e.g., "If it's a school day AND if you're not sick, you go to school. Otherwise, you stay home.").